Artificial Intelligence and Its Implications for Virginia's Workforce

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Abstract

Artificial intelligence (AI) has been on the horizon for years, but recent Census survey data indicates that it has arrived in Virginia workplaces. More businesses are incorporating it into their operations every day, and its impacts on the economy and the workforce are still taking shape. Measuring its adoption and effects is therefore a useful exercise for employers, workers, and policymakers alike.

Virginia Works has expanded on foundational research on workplace automation to evaluate how AI and other emerging technologies may influence labor markets across the Commonwealth. The study, conducted by Timothy Aylor, Virginia Works Senior Economist, introduces a framework for estimating AI adoption across work tasks, occupations, and industries. Recent advances in AI research have identified the specific tasks and skills AI can perform well. By linking these capabilities to job tasks, researchers can make informed assumptions about how AI may affect occupations, industries, and regional labor markets. In this context, AI "risk" represents the share of a job's tasks that AI could augment or replace.

Findings suggest the potential for widespread employment dislocation due to the adoption of artificial intelligence in the workplace. Technological change driven by AI will affect both non-routine and routine work tasks. Tasks that rely on social intelligence and creativity have so far been more resistant to AI, but even these boundaries are being pushed. While lower wage occupations were most affected by earlier waves of technological change, some higher paying fields such as computer and mathematical, legal, and financial occupations now have a significant share of job tasks with AI exposure.

Research on occupations and exposure to artificial intelligence has clarified who may be most affected by emerging technologies. It also highlights an important opportunity: individuals who invest in durable human skills can position themselves to benefit from the changes ahead.

AI is often discussed in terms of its potential negative impact on employment, but for many workers there are likely significant upsides, including greater productivity, job satisfaction, and higher wages. Even so, many workers will want to plan their careers in the age of AI or at least build adaptability as AI expands into more occupations. As seen in earlier waves of technology, the speed and extent of AI adoption will depend not only financial considerations, but also on broader societal and cultural factors. AI may overcome some of these barriers over time, but other influences, including the availability of capital and infrastructure, societal norms and regulations, may shape the pace of workplace automation.

Virginia is Leading in AI Adoption in the Workplace

April 2026 Census business survey data indicates that Virginia businesses have accelerated their adoption of AI and expect to expand its use even further. In fact, the District of Columbia has the highest rate of adoption, while Virginia is ranked fifth among states, following Washington, Colorado, and Arizona.

Several factors may explain this pattern. One possibility is cultural, with Virginia businesses and their workforces being more open to adopting AI than those in many other states. More likely, however, Virginia's industrial structure is a major driver of current and future workplace AI adoption. Nationwide, April Census business survey data shows that industries such as professional and business services have adopted AI more quickly than others, and these sectors make up a significant share of Virginia's economy. As a result, the mix of skills and job tasks in many of Virginia's leading industries may align more closely with what AI can currently do well. This makes intuitive sense, but is there a way to quantify this relationship?

A New Kind of Automation: 1.0 vs. 2.0

The conversation about artificial intelligence and work today sounds remarkably similar to the conversation about computerization that took place roughly 15 years ago. Both centered around powerful new technology, on the prospect of widespread displacement, and the concern that machines might outpace the workers who built them. The similarities are real, but so are the differences.

The earlier wave, Automation 1.0 relied on rule-based systems that followed predefined scripts and logic. These systems were task-specific and required human oversight. Automation 1.0 "impacted more lower-wage positions with limited educational background," particularly those involving routine manual or clerical tasks.

Automation 2.0, by contrast, is driven by artificial intelligence. AI systems employ machine learning and natural language processing, enabling them to adapt and improve over time. They are scalable, flexible, and capable of integrating across organizational functions. The research emphasizes that Automation 2.0 "impacts higher-wage positions with higher educational background," marking a significant departure from earlier patterns of technological disruption. This conceptual distinction is essential for understanding why AI affects a different set of occupations than previous automation waves.

AUTOMATION 1.0

- **Rule-based:** Following predefined scripts and logic
- **Task-Specific:** Automates individual repetitive tasks
- **Human Dependent:** Requires manual oversight
- **Impacted more lower-wage positions with limited educational background**

AUTOMATION 2.0

- **AI powered:** Uses machine learning and natural language processing
- **Self-Learning:** Adapts and improves over time with data
- **Scalable and Flexible:** Easily integrates across systems and functions
- **Impacts more high-wage positions with higher educational background**

How AI Exposure Is Measured

The exposure estimates in this report are built from the bottom up, at the task level, and then aggregated to the occupation, industry, and region. The approach borrows from the methodology pioneered by Frey and Osborne in their 2013 Oxford paper on computerization and refined by Eloundou and colleagues in their 2023 study of large language model exposure. The framework is straightforward in concept and rigorous in application.

1. Break an occupation into its tasks:	Use O*NET task statements to decompose each occupation into its constituent activities.
2. Assign an AI risk factor to each task:	Score each task based on how readily current AI tools can perform or meaningfully assist with it.
3. Aggregate task scores into an occupational AI risk score:	Weigh task-level scores to produce an overall exposure estimate for each occupational code.
4. Map occupational codes to regional employment:	Combine occupation exposure with regional industry prevalence to produce a geographic heat map.

Note on terminology: In this report, AI “risk” refers to the share of an occupation’s tasks that AI may either augment or displace. The term is not synonymous with job loss. A worker in a high-exposure occupation may experience higher productivity, expanded scope, and higher wages, or may see specific tasks delegated to AI tools while the role itself endures. Exposure is a measure of contact between AI and work, not a forecast of unemployment.

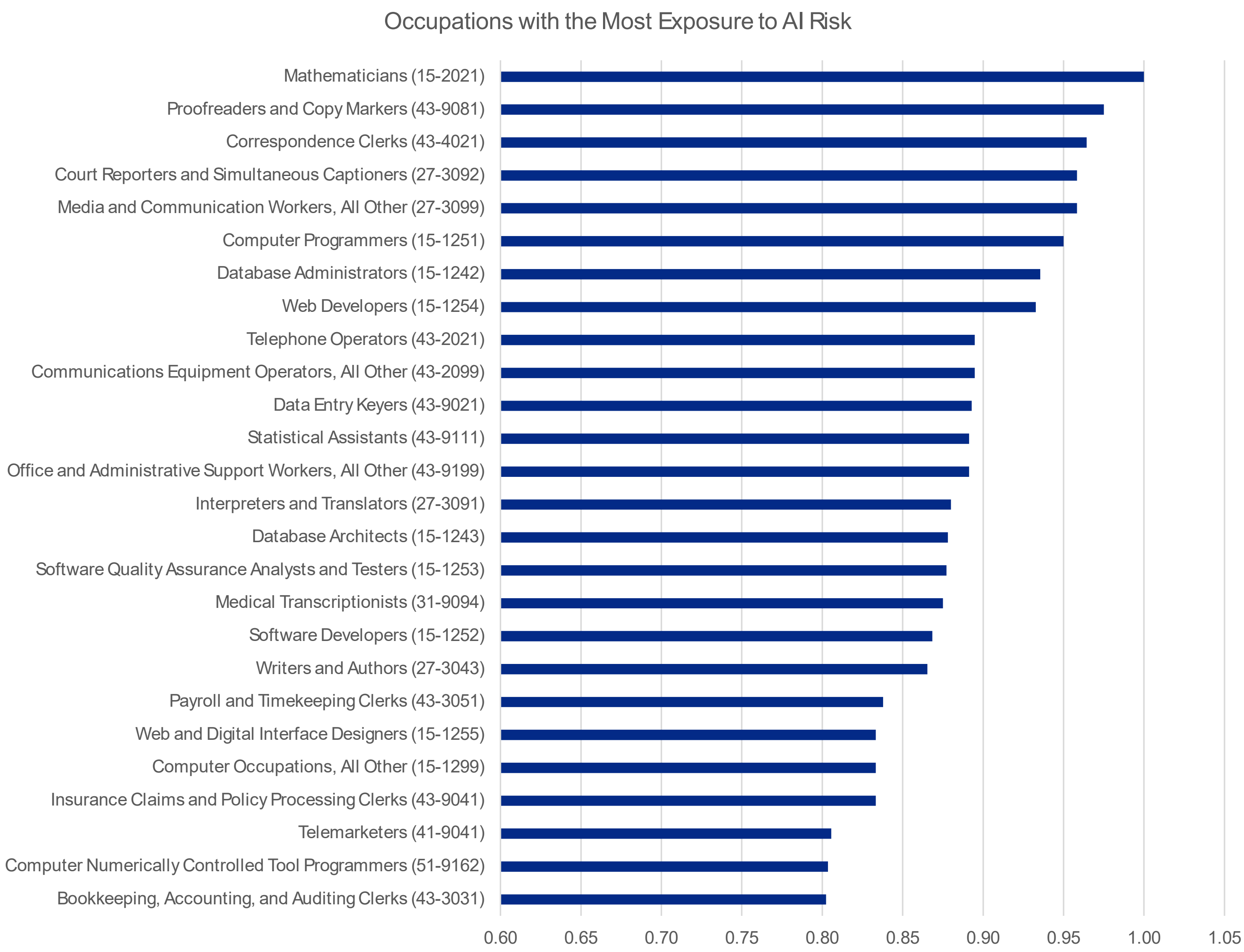
Occupational Findings

Occupational differences in estimated AI risk are driven by the key tasks that make up a job and how likely those will be impacted by AI. Recent research has quantified what tasks AI is good at performing and connected these strengths to specific job tasks. Medical transcriptionists, for example, illustrate how AI’s ability to successfully handle tasks like referencing, compiling, reviewing, proof reading, and even writing may augment, or potentially displace, this occupation in the future. AI performs well at clerical tasks that require meticulous attention to detail but little human judgment, like data entry and organizing documents. AI-powered accounting software can automate ledger entries, reconciliations, and basic financial reporting. Chatbots and virtual assistants can handle a high volume of routine inquiries, frequently asked questions, and basic troubleshooting. Many tasks performed by proofreaders, interpreters, and medical transcriptionists involve documenting, processing, and organizing information, making these roles particularly well-positioned to be augmented by AI and other emerging technologies.

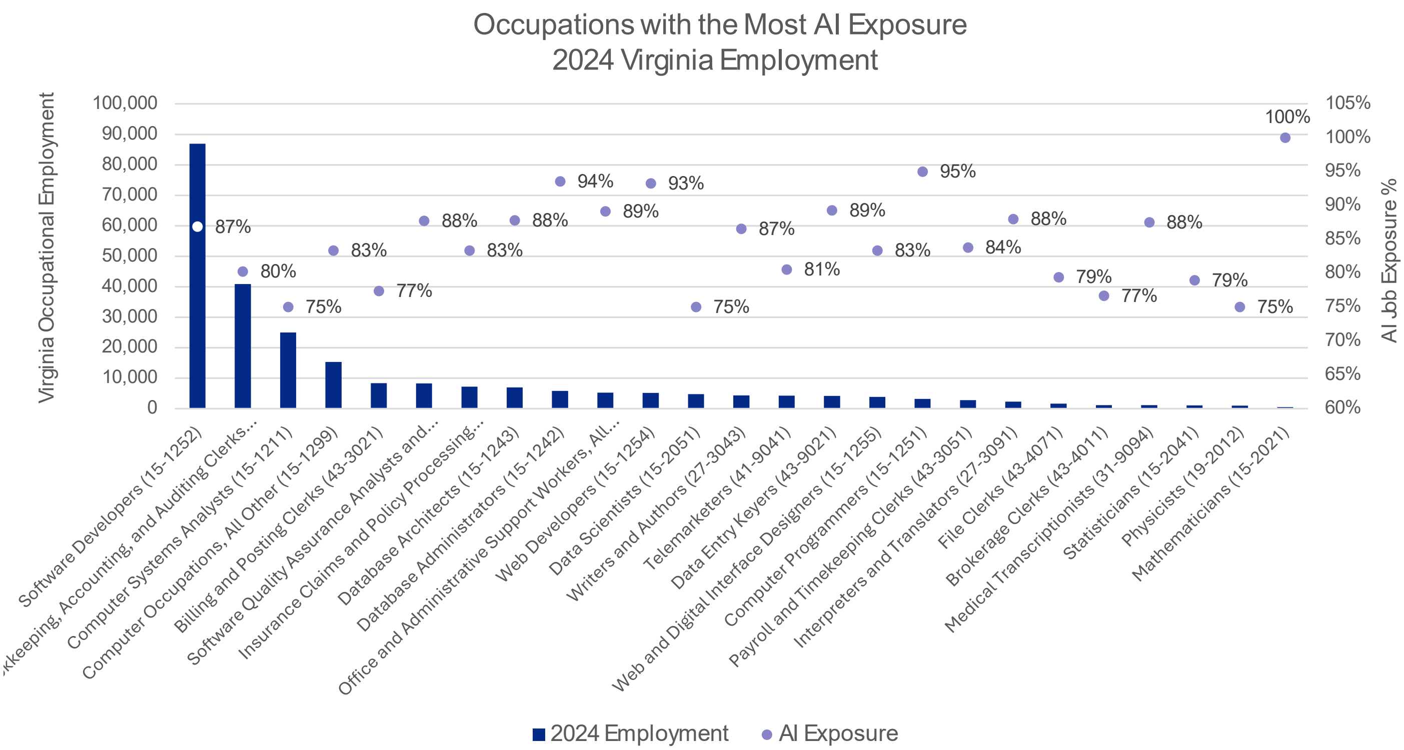
People bring different strengths to a job, and the mix of tasks can vary a lot between jobs in the same occupation. Looking at an occupation with moderate risk of AI exposure, such as sales representatives, shows that some job tasks are relatively shielded from AI encroachment, while others have significant exposure to AI, even though they are part of the same occupation. For example, an outdoor, hands-on product demonstration probably has less AI adoption risk, whereas robotics automate some physical tasks, jobs requiring high dexterity, adaptability to unpredictable environments, and common sense remain less susceptible. Similarly, promoting services and networking at a conference relies heavily on interpersonal and relationship building skills, abilities that are currently less likely to be replaced by technology.

However, a sales representative may develop marketing materials, maintain customer records, or calculate costs of goods and services. Although AI can produce marketing copy, social media posts, and news articles, creating imaginative or complex content is harder for AI to generate.

Taking these factors into account, the following chart highlights the Virginia occupations with the highest percentage of tasks that could be augmented or displaced by artificial intelligence. Some of these roles reflect a blend of impacts from both current AI capabilities and earlier waves of technology. For example, correspondence clerks and telephone operators may experience new AI-driven task displacement layered on top of earlier transitions brought about by email systems and digital telephony features such as voicemail and call-forwarding.



Taking the above raw scores for AI risk and putting them in context with actual employment across Virginia gives us a better understanding of the potential impacts. As an example, although mathematicians have the highest AI risk score, they have low employment across Virginia. Whereas software developers have a higher AI risk score as well as high employment which would indicate a greater impact to the Commonwealth.

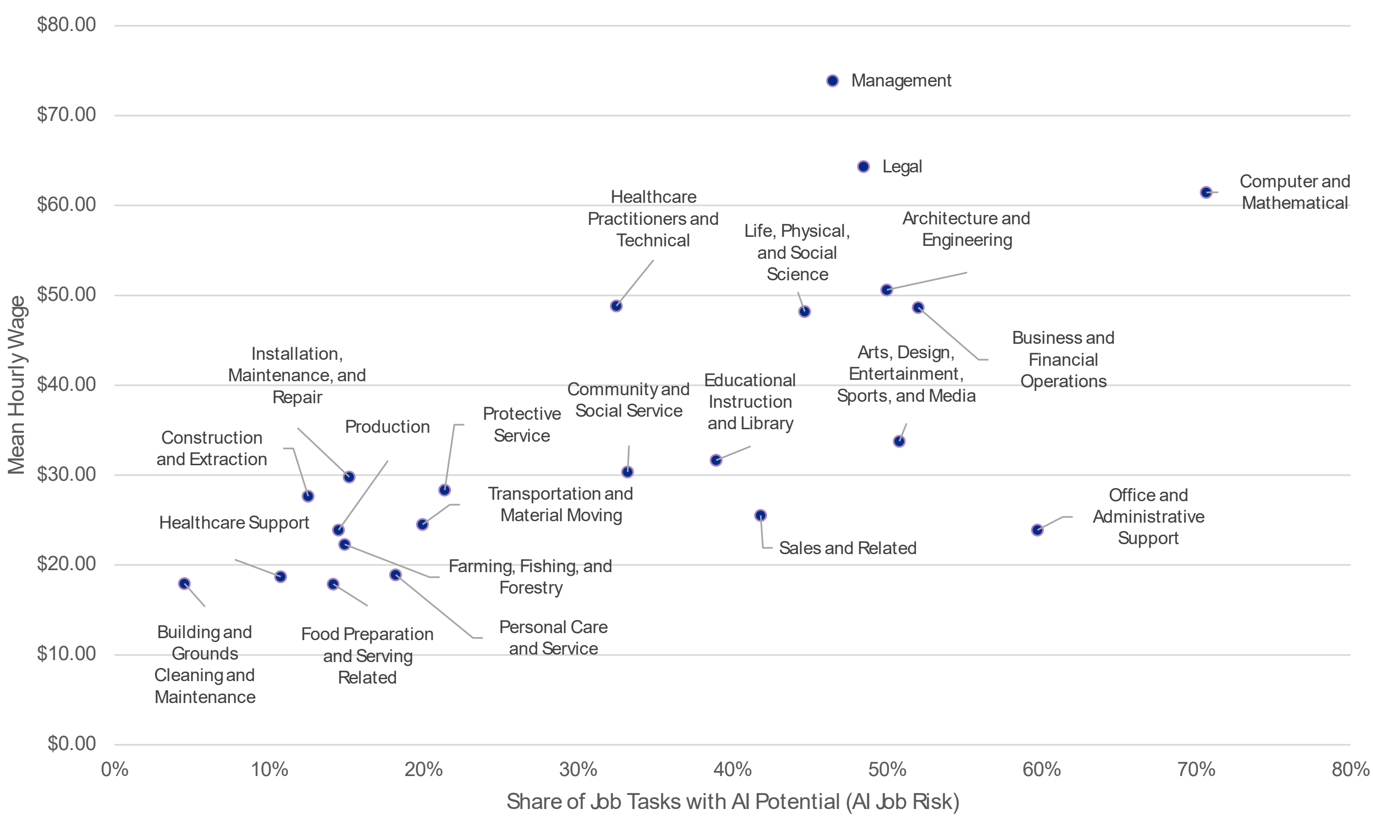


Wages and Educational Attainment

Many occupations with tasks suitable for AI adoption have higher than average wages as well as higher educational attainment. While lower wage occupations were most susceptible to earlier waves of technological change, occupations like computer and mathematical, legal, and business and financial occupations have significant shares of job tasks suitable for AI use. Average hourly wages of some of these include Management ($73.90), Legal ($64.35), and Computer and Mathematical ($61.48). Many of these types of jobs require post-secondary education. Others are administrative or customer support that do not require a post-secondary degree, but workers often have completed some college, moderate on-the-job training.

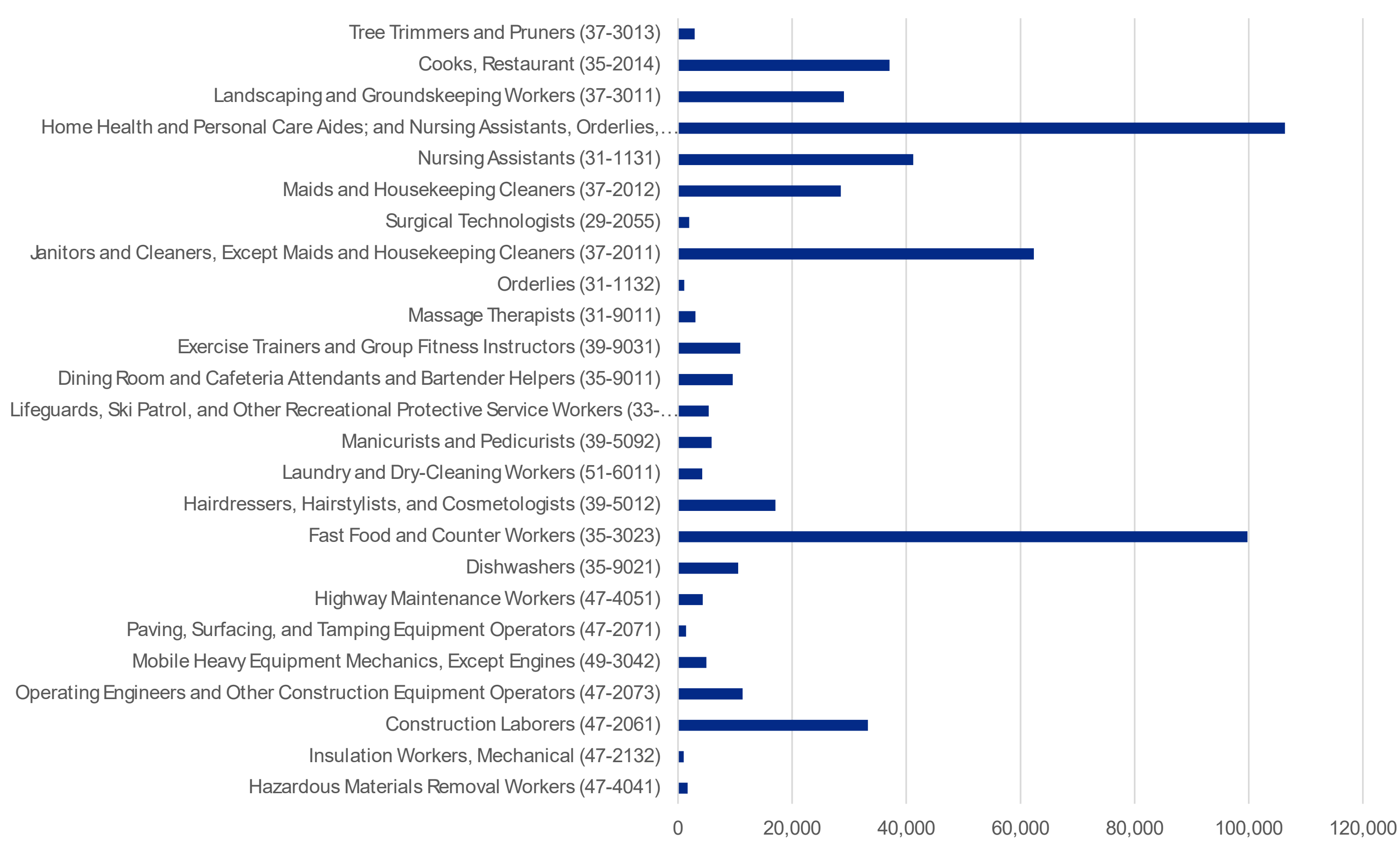
Occupational groups on the low end of the AI exposure scale include Food Production and Serving Related ($17.87), Building and Grounds Cleaning and Maintenance ($17.92) and Healthcare Support ($18.71). These often require more physical work, with less education and training required to start in those jobs.

Average Occupational Hourly Wage Compared with AI Exposure Share



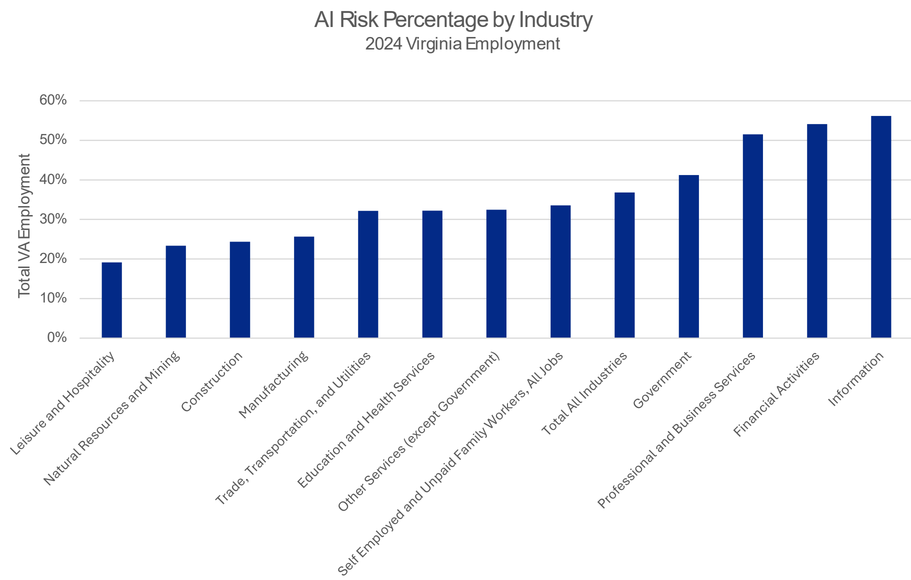
In-Demand Occupations with Low AI Exposure

2024 Virginia Employment



Industry Exposure in Virginia

Industries inherit the exposure of the workers they employ. An industry that draws heavily on software developers, financial analysts, paralegals, and administrative support staff will register higher exposure regardless of what the industry itself produces. An industry that draws on truck drivers, line cooks, construction and skilled trades will register low exposure for the same reason.



Why This Matters for Virginia

Virginia's economy is unique in its mix of industries with very different levels of AI exposure. Northern Virginia's concentration of professional services, federal contractors, and technology firms places it among the regions most affected by rapid advances in AI. At the same time, the Commonwealth is anchored by sectors with far lower automation risk such as manufacturing in Southwest Virginia and shipbuilding, port operations, and hospitality in Hampton Roads. This diversity is a strength, but it also means Virginia must take a deliberate, region-specific approach to AI adoption. To remain competitive, the state will need to lead in responsible AI integration while building adaptable workforce strategies that help workers and employers navigate technological change across all sectors.

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Industries that rely on workers in high AI-adoption occupations are likely to have greater AI exposure. For example, the Professional, Scientific, and Technical Services sector employs large numbers of software developers, bookkeepers, and computer systems analysts. AI risk for industries can be estimated by examining the mix of occupations required to be fully staffed. If an industry, like Financial Activities, has a lot of occupations with AI exposure, then that industry may be thought to be at risk as well. At-risk occupations in the Financial Activities sector include insurance claims and policy processing clerks, software developers, bookkeeping and accounting staff, auditing clerks, and administrative support roles.

These findings shed light on why Virginia may be one of the state leaders on industry AI adoption. Virginia's economy is driven by a unique mix of industries, many of which are likely to incorporate AI or plan to. In the private sector, this is led by the Professional, Scientific, and Technical Services sector, which employs approximately half a million people in Virginia. Banking and management of companies are additional private sector examples. Federal agencies involved in the administration of economic programs, state government universities and hospitals, and local government K-12 schools are government examples of important Virginia employers with potential AI exposure.

Looking Forward: Durable Skills and Human Centered Work

Although it may be tempting to read AI exposure scores as forecasts for job loss, the historical record cautions against interpreting as such. Every prior wave of technological shift, electrification, the personal computer, and the internet, generated both displacement and new categories of work. The early evidence on AI suggests a similar pattern, with one notable difference: the productivity enhancing effects are emerging far more quickly and penetrating more deeply into day to day tasks than previous transitions. AI may overcome these things over time, but other factors including availability of capital and infrastructure, societal norms and regulations may influence the pace of workplace automation using AI. Differences in these factors are projected to affect the pace of change across countries as well as U.S. regions.

The anticipated impacts of AI on the workplace will not be 'All or Nothing.' Many impacted workers may benefit from AI. For some workers, AI will provide productivity enhancement. AI used by workers or teams to increase output, automating or assisting with lower value tasks and enabling workers to spend more time on higher value tasks that are human centric. AI job risk shows up when enough tasks in a job are automated that the job itself becomes less needed.

Future Skills of the AI-Era Worker

As automation accelerates in the AI era, the nature of work will increasingly center on uniquely human skills that machines cannot replicate, those grounded in empathy, communication, critical thinking, adaptability, ethical judgment. As routine tasks become more automated, these distinctly human strengths will become even more valuable, essentially shaping how organizations create value and how workers remain resilient in an AI-enabled economy.

The future ready worker in Virgina will understand what AI can and cannot do, how to use it safely, and how to evaluate its outputs. They will also bring meaningful industry, customer, regulatory, or craft knowledge that gives AI purpose. Above all, they will apply durable human skills that transform AI-augmented output into real economic and social value.

Strengthening AI knowledge and skills is a sound strategy for nearly any professional, providing a pathway to greater career stability while laying the foundation for long-term growth in an increasingly technology-driven economy. Virginia Works is leading the way by providing resources that help residents build the knowledge and skills needed to use AI effectively in their professions. Free online training is available to Virginia residents to want to learn AI fundamentals to boost productivity, deepen their AI expertise, or gain advanced skills to become leaders in the field. Professional certificates and hands-on training is additionally offered through Virginia Works' partnership with Google. The Grow with Google program provides introductory and advanced learning pathways, along with real world practice, to help learners build confidence in applying AI tools in professional settings.

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